

# Human-Centered Explainable Vision System

Technical Report — Project Documentation

Vaishnavi Marihal | May 2026

B.E. Computer Science, Jain College of Engineering and Technology, Hubballi

## 1. Motivation

Standard image classifiers return a label and a confidence score. These outputs are statistically informative but interactively insufficient. A user receiving the output "golden retriever: 81.5%" cannot determine which visual features drove the prediction, whether the confidence score reflects genuine certainty or distributional ambiguity, what alternative interpretations the model considered, or whether the prediction is trustworthy for their specific context.

This gap between statistical correctness and human usability motivated this project. The central question is not how to make the classifier more accurate, but how to make its outputs more interpretable to a human operator.

This concern emerged directly from prior project work. A fake news detection system achieving 98% accuracy returned binary predictions with no mechanism for the user to understand why a particular article was flagged or how confident the system actually was. A YOLOv4 video summarisation system detected human activity with measurable precision but produced clips with no contextual framing for the operator. In both cases, the system was statistically competent and interactively hollow. This project is a first attempt to address that hollowness systematically.

## 2. System Design

The system is structured around five components: a visual classifier, an uncertainty module, a saliency map generator, a query interpreter, and an explanation engine. The design is deliberately constrained. Interaction is structured through five predefined explanation intents rather than open-ended natural language. This constraint is intentional — a more interpretable interaction architecture for a project about interpretability.

### 2.1 Pipeline

```
Image upload → ResNet50 inference (top-5 predictions + full softmax) →  
Entropy computation + confidence band assignment → Grad-CAM heatmap  
generation → User query → Rule-based intent detection → Explanation engine  
→ Visual + textual response
```

### 2.2 Classifier

A pretrained ResNet50 from torchvision is used for inference only — no fine-tuning was performed. ResNet50 was selected for its well-documented layer structure, which makes Grad-CAM hook registration straightforward, and for its strong ImageNet baseline performance. The model returns the full 1000-class softmax probability distribution, which is used downstream for entropy computation.

### 2.3 Uncertainty Module

Uncertainty is quantified via Shannon entropy over the full softmax distribution:

$$H(p) = -\sum p_i \log(p_i) \text{ for } i = 1 \text{ to } 1000$$

Raw entropy is normalised against the maximum possible entropy for 1000 classes ( $\log 1000$ ), yielding a value in  $[0, 1]$ . A normalised entropy of 0 indicates a perfectly certain prediction. A value

of 1 indicates maximal uncertainty with probability spread equally across all 1000 classes.

Three confidence bands are defined by entropy thresholds:

| Band      | Entropy Range | Interpretation                          |
|-----------|---------------|-----------------------------------------|
| High      | < 0.30        | Probability concentrated on few classes |
| Cautious  | 0.30 – 0.55   | Moderate spread across classes          |
| Ambiguous | > 0.55        | High distributional uncertainty         |

Effective class count is computed as  $\exp(H)$  — the number of classes carrying meaningful probability mass. A value of 2.4 indicates the model is effectively choosing between approximately 2-3 classes, regardless of which specific classes those are.

## 2.4 Grad-CAM

Grad-CAM is implemented from first principles following Selvaraju et al. (2017), without any external explainability library. Forward and backward hooks are registered on ResNet50's final convolutional layer (layer4). For a target class  $c$ :

$$\alpha_k^c = (1/Z) \sum_{ij} (\partial y^c / \partial A_{ij}^k) L_{\text{GradCAM}} = \text{ReLU}(\sum_k \alpha_k^c \cdot A^k)$$

Where  $A^k$  is feature map  $k$  from layer4,  $y^c$  is the pre-softmax score for class  $c$ , and  $\alpha$  represents the global average pooled gradient — the importance weight for each feature map. ReLU retains only regions that positively contributed to the target class prediction. The resulting heatmap is upsampled to 224x224 and overlaid on the original image using a JET colormap.

## 2.5 Query Interpreter

User queries are mapped to one of five structured intents using rule-based regex pattern matching. No NLP library or transformer model is used. This design decision is intentional — using a black-box model to parse queries in a system designed to explain black-box predictions would be architecturally contradictory.

| Intent                 | Example Query                 | Output                             |
|------------------------|-------------------------------|------------------------------------|
| why_prediction         | "Why did you classify this?"  | Grad-CAM + attention description   |
| confidence             | "How confident are you?"      | Entropy + band + effective classes |
| important_region       | "Which part mattered most?"   | Grad-CAM + spatial region label    |
| alternative_prediction | "What else could this be?"    | Top-3 alternatives + overlap note  |
| comparison             | "Why this instead of second?" | Side-by-side Grad-CAM for top-2    |

## 3. Evaluation

25 images were evaluated across five categories of increasing visual complexity. For each image, top-5 predictions, normalised entropy, confidence band, Grad-CAM focus pattern, and top-2 probability overlap were logged. Full results are available in [evaluation/evaluation\\_log.csv](#).

| Category  | N | Avg Entropy | Avg Top-1 Prob | Diffuse Heatmaps |
|-----------|---|-------------|----------------|------------------|
| Clear     | 3 | 0.0939      | 0.7905         | 1/3              |
| Moderate  | 5 | 0.2324      | 0.5693         | 2/5              |
| Cluttered | 5 | 0.3658      | 0.5217         | 5/5              |

|         |   |        |        |     |
|---------|---|--------|--------|-----|
| Similar | 5 | 0.0652 | 0.8281 | 4/5 |
| OOD     | 4 | 0.4207 | 0.3414 | 1/4 |

### 3.1 Key Observations

**O1 — Entropy tracks visual complexity monotonically.** Normalised entropy increased consistently from clear (0.0939) through moderate (0.2324) to cluttered (0.3658) and OOD (0.4207). Entropy functions as a reliable proxy for image-level uncertainty across varied scene types.

**O2 — Cluttered images universally produced diffuse Grad-CAM activation.** All 5 cluttered images produced diffuse heatmaps. When scenes contain multiple objects or busy backgrounds, the model cannot localise attention — activation spreads broadly. Visual complexity degrades spatial interpretability alongside confidence.

**O3 — Confidence and attention focus are partially independent.** Several high-confidence predictions produced diffuse Grad-CAM maps. Entropy-based confidence and Grad-CAM spatial focus capture different properties of model behaviour. A system reporting only confidence scores would miss this distinction.

**O4 — Inter-class similar images showed unexpectedly low entropy.** The similar category produced average entropy of 0.0652 — lower than clear images. The model commits decisively even on visually confusable pairs. Low entropy reflects model commitment, not necessarily correctness.

**O5 — OOD images correctly triggered high uncertainty.** Out-of-distribution inputs produced the highest average entropy (0.4207) and lowest top-1 probability (0.3414). The uncertainty module correctly flagged these with cautious or ambiguous bands.

**O6 — One clear-category image misclassified with moderate confidence.** A red apple was classified as strawberry (37.5%, high band). Confidence bands reflect distributional certainty, not ground-truth correctness — a known limitation of softmax-based uncertainty estimation.

## 4. Limitations

- **Grad-CAM explanations are correlational, not causal.** Highlighted regions influenced the prediction within the model's learned representation. They do not constitute a causal account of class membership.
- **Rule-based interaction limits linguistic flexibility.** Queries outside the five intent patterns return an unknown response. This is a deliberate design constraint, not an oversight.
- **Softmax confidence is not perfectly calibrated uncertainty.** ResNet50 is known to be overconfident. Normalised entropy provides a better signal than raw top-1 probability but remains derived from an uncalibrated distribution.
- **Explanations inherit ImageNet biases.** Predictions and activations reflect ImageNet class boundaries, which may not align with human conceptual categories.
- **Evaluation is qualitative and small-scale.** 25 images across informal categories. Findings are observational, not statistically validated.

## 5. Future Work

- Calibrated uncertainty via temperature scaling or Monte Carlo Dropout to address softmax overconfidence
- User-adaptive explanation verbosity based on inferred operator expertise
- Multimodal interaction allowing image region selection alongside text queries
- LLM-assisted explanation refinement as a post-processing layer over structured engine outputs

## 6. Reference

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Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. *ICCV 2017*.

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*This report documents system design, methodology, evaluation findings, and limitations of the Human-Centered Explainable Vision System. Full source code and evaluation data available at [github.com/Vaishnavi-Marihal/human-centered-explainable-vision](https://github.com/Vaishnavi-Marihal/human-centered-explainable-vision)*